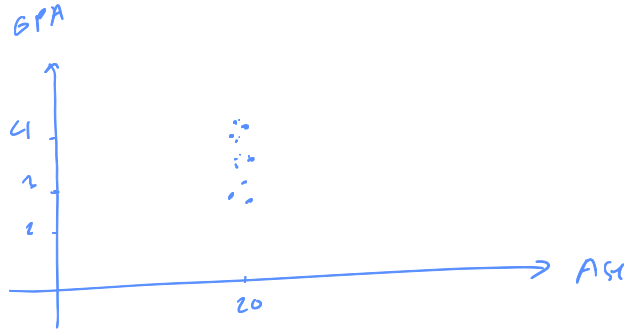
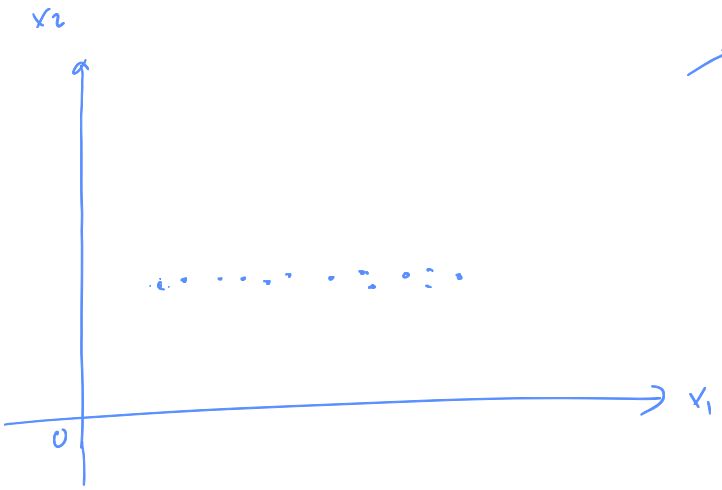
Age	GPA
20	3.5
20	2.1
20	4.0
20.5	3.9
20	1.6
20	2.9
19.5	3.5

① GPA contains more information than Age.

② Age is almost a constant, does not contain much information.

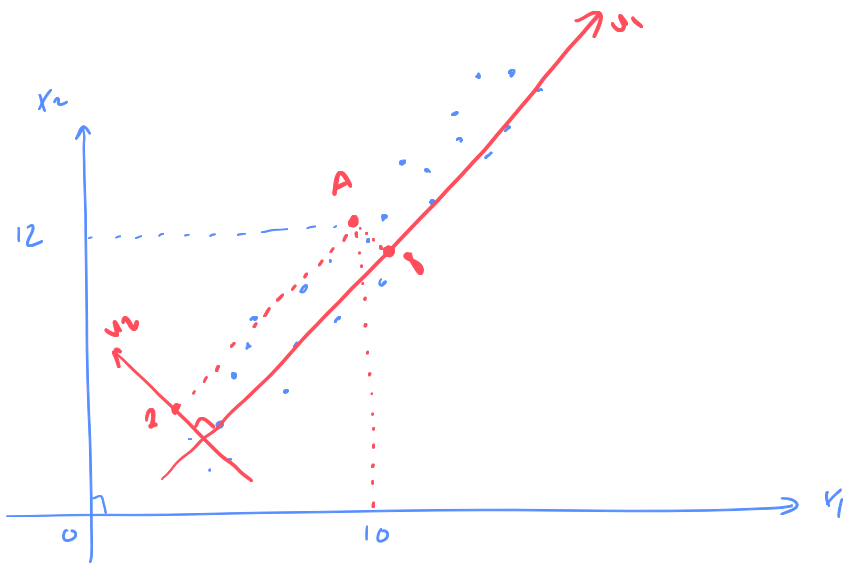
③ If we want to remove a variable, we would remove "Age"





① X_2 does not contain much information

② If we want to remove a variable, we would remove X_2



	X_1	X_2		U_1	U_2
(A)	10	12	→	8	2

U_1: The first principal component

U_2: The second principal component.

we can remove U_2 if we want the data has only 1 variable.

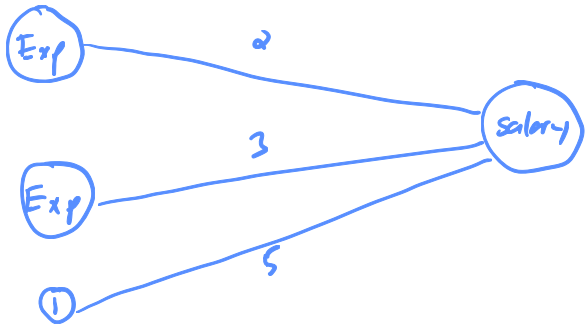
Neural Network

$$Y = 2x_1 + 3x_2 + 5 \quad \text{(linear model)}$$

Y → Salary

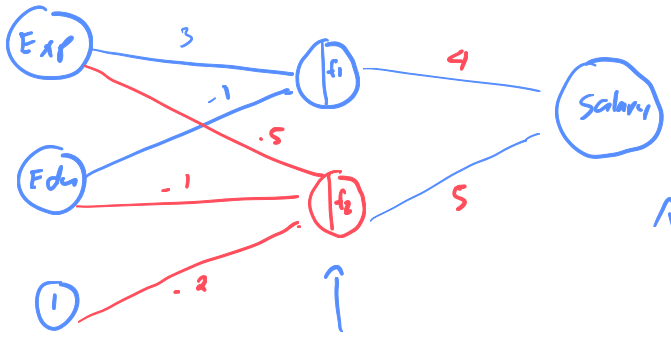
x_1 → Experience (Exp)

x_2 → years of education (Edu)



(a neural network)

Model 2



hidden layer

output layer

Input layer

f_1 ; f_2 are "activation" functions

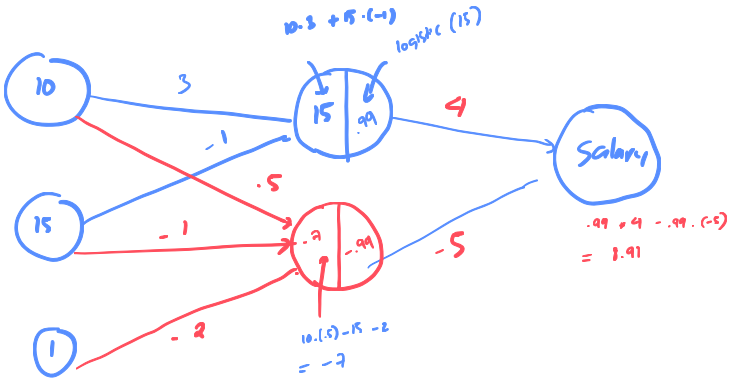
f_1 : logistic function

f_2 : tanh function.

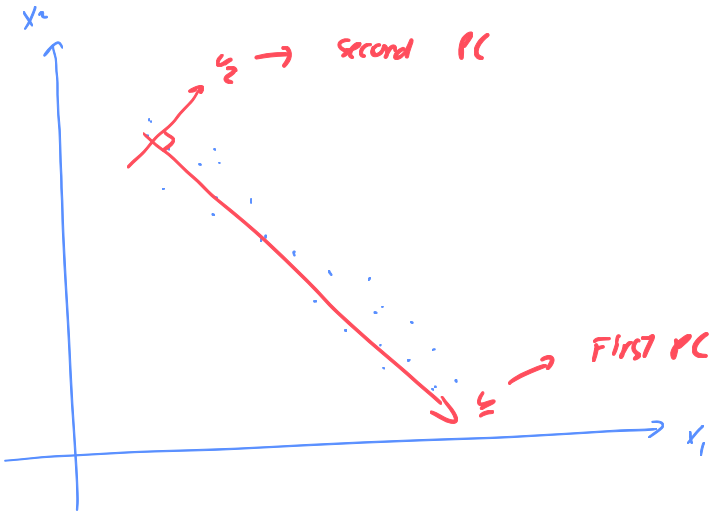
Calculation example

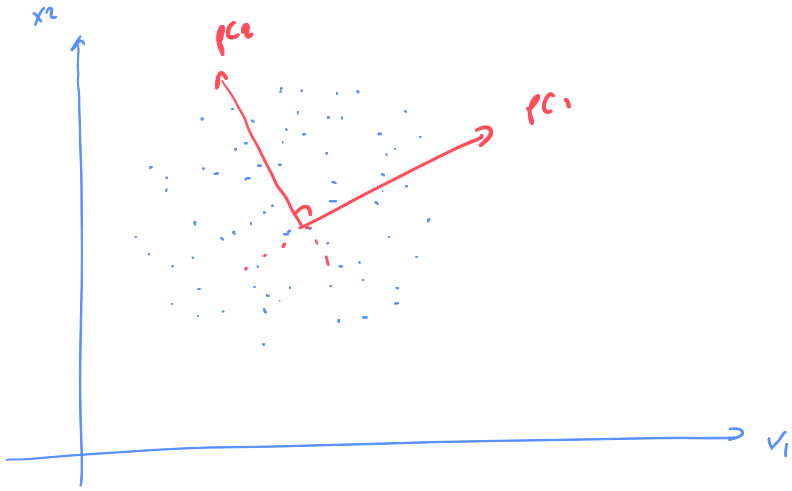
Exp = 10 years

Edu = 16



CLASS EVALUATION





In this case, PCA does not really help.